

X 2. A TCP sender has $CWND = 8$ *0/1
MSS and $SSTHRESH = 16$. After
one RTT with no loss, $CWND$
becomes:

- ☐ a) 9 MSS
- ☒ b) 16 MSS
- ☐ c) 12 MSS
- ☐ d) 8 MSS

Correct answer

- ☒ a) 9 MSS

*0/1

13
20

X

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65%

✓ 12. HTTP/2 multiplexing solves *1/1
HOL blocking at the HTTP layer
but not at the:

- ☐ a) Application layer
- ☐ b) DNS layer
- ☒ c) TCP layer
- ☐ d) IP layer

✓

? X 9. QUIC protocol uses UDP as its *0/1
transport. The key advantage over

✗ 9. QUIC protocol uses UDP as its transport. The key advantage over TCP-based TLS is: *0/1

- ☐ a) Eliminating IP header overhead
- ☐ b) 0-RTT or 1-RTT connection setup
- ☒ c) Removing encryption ✗
- ☐ d) Native multicast support

Correct answer

- ☒ b) 0-RTT or 1-RTT connection setup

✓ 15. Which DNS record type maps a domain name to an IPv6 address? *1/1

- ☐ a) A
- ☐ b) CNAME
- ☒ c) AAAA ✓
- ☐ d) MX

✓ 17. Which attack exploits the UDP-based DNS protocol by sending forged responses with a matching transaction ID? *1/1

?

CSA0720-MCQ

Total points 13/20 ?

C03-AT3

✗ 16. A DNS resolver switches from *0/1
UDP to TCP when:

- ☐ a) The query type is MX
- ☐ b) The response is truncated (TC bit set)
- ☒ c) TTL exceeds 3600 s ✗
- ☐ d) The authoritative server is unreachable

Correct answer

- ☒ b) The response is truncated (TC bit set)

✓ 8. What is the maximum UDP *1/1
payload size without IP-level
fragmentation on a standard
Ethernet link (MTU = 1500 bytes)?

- ☐ a) 1480 bytes
- ☒ b) 1472 bytes ✓
- ☐ c) 1460 bytes

✗ 10. HTTP/1.1 persistent connections reduce latency by: *0/1

- ☒ a) Enabling UDP fallback ✗
- ☐ b) Reusing the same TCP connection for multiple requests
- ☐ c) Compressing HTTP headers
- ☐ d) Sending requests without waiting for DNS

Correct answer

- ☒ b) Reusing the same TCP connection for multiple requests

✓ 5. In a high-latency satellite link (600 ms RTT), TCP throughput is primarily limited by: *1/1

- ☐ a) MTU size
- ☒ b) CWND growth rate relative to RTT ✓
- ☐ c) DNS resolution time
- ☐ d) UDP competition

② ✓ 18. Setting a very low TTL (e.g., 30 s) for a DNS record primarily *1/1

✓ 18. Setting a very low TTL (e.g., 30 s) for a DNS record primarily increases: *1/1

- ☐ a) Cache hit ratio
- ☒ b) Recursive resolver query load ✓
- ☐ c) DNSSEC signature validity
- ☐ d) Zone transfer frequency

✓ 3. Which field in the TCP header is used to implement flow control at the receiver side? *1/1

- ☐ a) Sequence number
- ☐ b) Acknowledgement number
- ☒ c) Window size ✓
- ☐ d) Urgent pointer

✗ 13. Which HTTP status code indicates that a resource has permanently moved and browsers should update bookmarks? *0/1

?

- ☐ a) 301

✓ 17. Which attack exploits the UDP-based DNS protocol by sending forged responses with a matching transaction ID? *1/1

- ☐ a) ARP poisoning
- ☒ b) DNS cache poisoning ✓
- ☐ c) SYN flooding
- ☐ d) BGP hijacking

✓ 1. Which TCP mechanism reduces the congestion window by half upon detecting packet loss via duplicate ACKs? *1/1

- ☐ a) Slow Start
- ☐ b) Fast Retransmit
- ☒ c) Fast Recovery ✓
- ☐ d) Tahoe Reset

✓ 6. UDP is preferred over TCP for DNS queries primarily because: *1/1

?

- ☐ a) UDP supports fragmentation

✗ 7.A UDP-based streaming application experiences 15% packet loss. The best application-layer mitigation is: *0/1

- ☐ a) Increase UDP buffer size
- ☒ b) Switch to TCP ✗
- ☐ c) Implement Forward Error Correction (FEC)
- ☐ d) Reduce frame rate to 1 fps

Correct answer

- ☒ c) Implement Forward Error Correction (FEC)

✗ 14. In HTTP caching, the Cache-Control: no-cache directive means: *0/1

- ☐ a) Never cache this resource
- ☐ b) Validate with the server before using a cached copy
- ☒ c) Cache for 0 seconds ✗
- ☐ d) Disable all caching headers

Correct answer

- ☒ b) Validate with the server before using a cached copy

✓ 4. Nagle's algorithm causes increased latency primarily because it: *1/1

- ☐ a) Disables ACK piggybacking
- ☒ b) Buffers small segments until ACK arrives ✓
- ☐ c) Reduces CWND on every packet
- ☐ d) Increases SYN backlog

✓ 19. A user browses to anneww website. The correct sequence of protocol interactions for the first byte of the page is: *1/1

- ☐ a) HTTP, TCP, DNS
- ☒ b) DNS, TCP handshake, HTTP GET ✓
- ☐ c) TCP, DNS, HTTP
- ☐ d) DNS, HTTP, TCP

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✓ 20. Under 20% packet loss, which protocol combination gives the best performance for a real-time multiplayer game? *1/1

☐ a) TCP for game state, HTTP for assets

☒ b) UDP with application-layer sequencing for game state, HTTP/3 for assets ✓

☐ c) HTTP/2 for all traffic

☐ d) TCP with Nagle enabled for all traffic

✓ 11. Head-of-line blocking in HTTP/1.1 occurs when: *1/1

☐ a) The DNS server is slow

☒ b) A response blocks subsequent responses on the same connection ✓

☐ c) TCP CWND drops to zero

☐ d) The server rejects pipelined requests

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